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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-DS-010

1. Topic

Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Caregiving has become challenging due to the aging population and chronic illnesses exposing caregivers under extreme physical and psychological stress. According to research, increasing workloads are linked with elevated anxiety and depression [1], [2]. For caretakers of elderly people who are deaf, this stress is significantly greater. In addition to severe communication barriers 70% of people over 70 have hearing loss, and only 7% have access to interpreters [4] new caregivers need to cope with serious unpreparedness. According to studies, "limited caregiving knowledge" and "lack of hospital training" are significant obstacles that leave caregivers without medical backgrounds feeling overburdened [5].

This knowledge gap is made worse by sign language attrition; without frequent training, caretakers forget key signs, resulting in serious gaps during treatment or emergencies. Despite these demands, caregiver mental health is not tracked and there are no predictive stress models, which can result in silent burnout [3] and affect care recipients safety when physiological stress differs from personal feelings [2]. The main users are overburdened nurses and caregivers who must deal with the triple burden of unpreparedness, staffing shortages, and communication barriers with deaf care recipients [4], [5].

Impact:

- Social: Reduced well-being lowers quality of life and quality of care [1], [3].
- Technical: Integrated stress prediction and skill support are absent from the current segregated tools (wearable vs. survey) [2].

- Economic: Expensive absenteeism and turnover are caused by burnout.
- Legal/Ethical: Inadequate training violates the duty of care, and neglecting access to communication violates the rights of people with disabilities.
- High staff turnover causes resource inefficiencies.

The Center of Excellence for AI (CoEAI) in Digital Health and Smart Wellness is the FoC Research Cluster, aligned with SDG 3 (Good Health and Well-being) through its focus on mental health monitoring and sustainable caregiving systems.

References

[1] M. Mazhar, H. Mahmood, and M. Speck, "Impact of caregiving burden on the mental health of caregivers: A study of individuals caring for chronically ill care recipients," J. Prof. & Appl. Psychol., vol. 5, no. 2, pp. 144–157, 2024. [Online]. Available: https://iprpk.com/ojs/index.php/jpap/article/view/269

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[3] C. Shi, X. Zhou, N. Zou, S. Chen, and J. Shi, "Predicting psychological distress in informal caregivers of older people with dementia: development and interpretation of a nomogram and XGBoost model," BMC Nursing, vol. 24, art. no. 1311, 2025, doi: 10.1186/s12912-025-03905-0.

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6. Brief description of Existing Research and Systems

Existing mental health monitoring systems rely on wearable devices like Fitbit and Apple Watch to track physiological indicators. Recent research shows that these methods are combined with machine learning models like Support Vector Machines (SVM) and Random Forest, they may reach up to 99.98% accuracy in diagnosing stress levels using EEG signals [1]. However, these approaches are correlation-based and lack the causal inference capabilities needed to discover the root causes of burnout. While consumer apps like Calm and Wysa offer conversational support, they often use static scripts or simple NLP, lacking the deep integration of a "Human Digital Twin" (HDT) required for tailored care.

HDT frameworks are emerging as a valuable tool for precision health, combining wearable data with behavioral patterns to model individual health patterns. Recent implementations of mental health digital twins employing EEG data have achieved classification accuracy of approximately 85.7% [2]. Despite this progress, voice-based activity logging in these systems remains in its early phases, with standardized inputs typically replacing natural language interactions. In terms of accessibility, technologies like NVIDIA's Signs provide real-time feedback to sign language learners via 3D avatars and computer vision [3]. Advanced research frameworks, such as SignAvatar, rely on transformer-based Conditional Variational Autoencoders to accurately reconstruct and produce 3D sign language movements [4]. However, these systems prioritize accuracy above verified memory or personalized learning adaptation. Multi-agent reinforcement learning shows possibilities for decision-making in the broader healthcare AI domain. The MedAgent-RL architecture, for example, uses a curriculum-learning-guided technique to optimize collaboration between AI agents, resulting in a 23.6% improvement in medical reasoning tasks when compared to standard baselines [5]. Despite these technological developments, a comprehensive solution is still missing. There is currently no solution that combines burnout causation graph discovery with voice-enabled digital twins and retention-focused 3D avatar learning. Addressing these fragmented challenges within a uniform, privacy-preserving framework creates a tremendous potential for innovation.

References

[1] V. Chandra, "Machine learning-based stress classification system using wearable sensor devices," *Int. J. Artif. Intell.*, vol. 13, no. 1, pp. 341, 2024. Available: https://ijai.iaescore.com/index.php/IJAI/article/view/22294

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7. Brief description of the solution's nature, including a conceptual diagram.

The proposed Caregiver Co-Pilot solution is a fully integrated mobile and web solution aimed at enabling caregivers to significantly alleviate their physical and cognitive burdens. The solution not only focuses on caring for the elderly but is specifically designed to assist the caregiver, filling the gap between high care requirements and scarce resources. The solution transforms caregiving from a reactive and stressful task to a proactive, manageable task.

Burnout Prevention: Addressing the gap in mental health support, the system utilizes Temporal Causal Analysis. Unlike traditional tracking tools, the system forecast the caregiver's burnout "regime shifts" and recommends healthy tips to recover before becoming exhausted.

Safety Monitoring: Due to the challenges posed by the burden of constant surveillance, the system employs the use of privacy preserving skeletal safety monitoring. This reduces the burden on the caregiver by identifying potential falls through anonymous joint data, thereby removing the need to constantly monitor the video feeds of the seniors.

Skill Enhancement: The AI-Powered sign language tutor to minimize the effects of communication gaps and enables caregivers to learn the skill of non-verbal communication easily. This is possible with the aid of explainable and real-time feedback provided by 3D avatars. This reduces the frustration experienced in communicating with seniors who use non-verbal communication.

Administrative Relief: To cut down upon the administrative tussle of paperwork, voice-powered documentation introduces the facility for caregivers to create well-structured shift reports just by speaking.

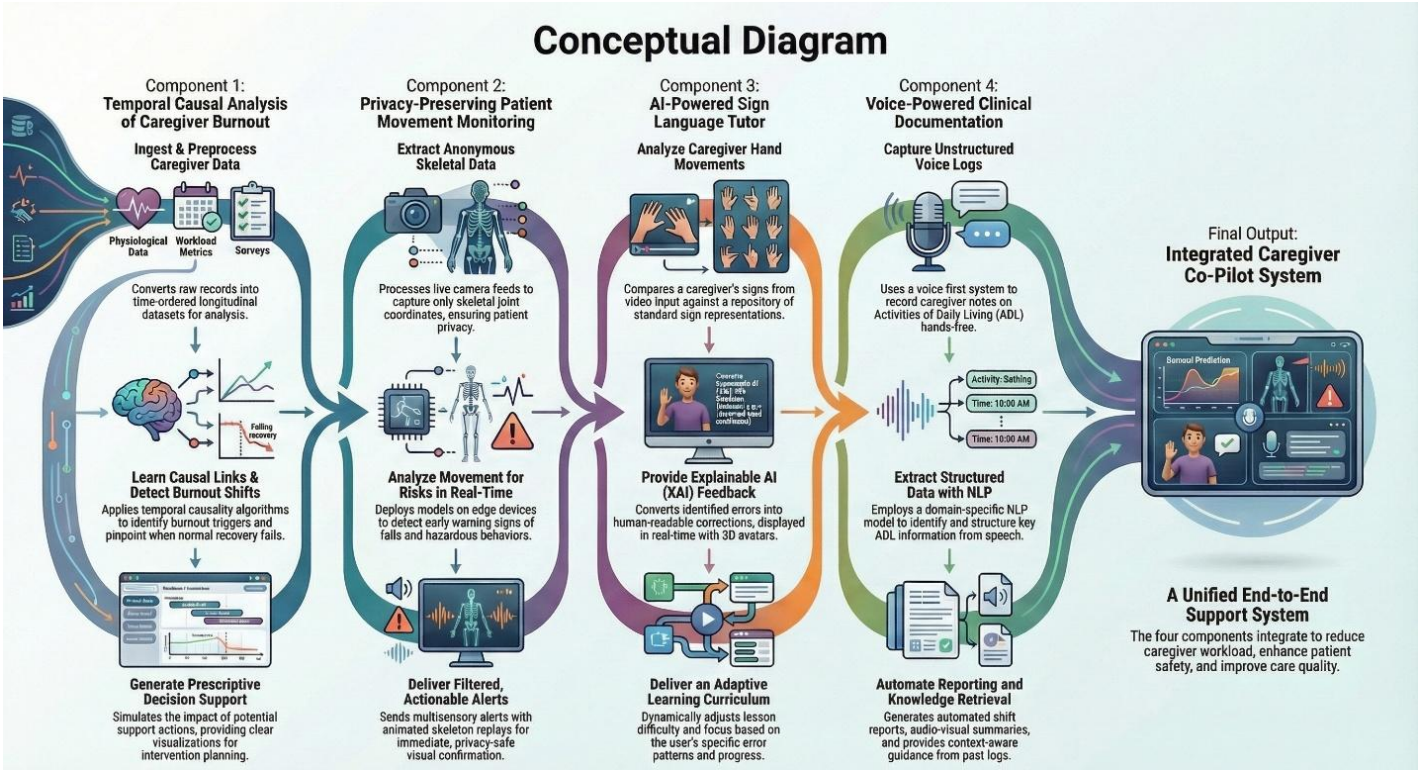
The system works by incorporating multimodal data sources such as the physiological and workload data of the caregiver, the privacy filtered video streams from the elder's room, hand gestures, and unstructured voice logs in English and Singlish. The data sources enable the operation of four customized processing engines aimed at catering to the caregiver.

The Causal Inference Engine evaluates the data trends to discover burnout causes and relief options, while Edge AI and Computer Vision process video locally to detect risks via anonymous skeletal movements, ensuring the elder's privacy is respected.

To further support the caregiver's daily tasks, Explainable AI and 3D rendering technologies compare their hand gestures against standard signs, generating adaptive lessons and immediate corrective feedback for skill improvement.

The Automatic Speech Recognition and Natural Language Processing algorithm allows the caregiver's spoken voice to be transcribed and translates verbalized notes automatically into a structured log format for organized activities of everyday living.

The final output will be an integrated mobile and web application that gives the caregiver tips to improve their stress and burnout, safety recommendations that are graphically depicted through stick-figure replays, skill-building recommendations that are interactive, and daily reports as dictated that are auto-generated and professional.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data.

In order to provide a technically sound and realistic solution that meets the practical requirements of care givers environment, we will incorporate a systematic plan for domain knowledge acquisition. The requirement engineering process is done through semi-structured interviews and shadowing activities, grouping the population of interest into trainee caregivers and senior caregivers. We thereby permit the detection of different behavioral patterns among distinct sub-populations: the learning difficulty of acquiring skills by raw trainees versus the cumulative burnout measures of experienced caregivers. This is useful work to embed ground-truth labels to build a causal model or to develop a basis to compare learning material to actual care standards.

Our team will be working on closing the theory-practice gap by reproducing best practices or benchmarks from the latest literature on State-of-the-Art Deep Learning models. In the area of workforce dynamics, we're moving beyond traditional predictive regression analysis and exploring the application area of Structural Causal Models (SCMs). We're building competence and skills on representing time-varying confounders through Directed Acyclic Graphs (DAGs) and Neuro-Symbolic computations for the inference of "what if" ("counterfactual") outcomes under stress intervention.

In movement analysis, our area of interest is Geometric Deep Learning on Non-Euclidean data. We're working on Spatiotemporal Graph Convolutional Networks (ST-GCNs) that combine spatial information from skeletal topologies with dynamics, which do not require the intensive processing characteristic of processing pixels. Finally, to contribute to the "Caregiver Co-Pilot" project, we're working on the adoption of Edge AI. We're working on Model Compression pipelines Post-Training Quantization and Pruning on Transformer Models in NLP optimized for fast on-device processing on mobile hardware limitations.

The data acquisition for the research relies on the latest data engineering principles and data augmentation techniques for developing robust models for the study while satisfying the constraints of privacy. To conduct causal analysis of caregiver burnout, temporal data are primarily acquired through longitudinal surveys, psychometric questionnaires, and co-teaching strategies within a deep learning framework. The co-teaching approach is adopted to mitigate noise in self-reported stress measures, which commonly arise due to subjective bias and variability in caregivers' stress perceptions.

For the computer vision and sign language tasks, we use a Transfer Learning strategy. We make use of publicly available datasets and pre-train our Feature Extractors on basic human actions. We then fine-tune these networks on a carefully selected and privacy-compliant dataset for skeletal coordinates. Training on vectorized data instead of pixels and complementing this with synthetically generated data reduces the requirement for sensitive video data while preserving high model accuracy through domain adaptation.

9. Objectives and Novelty

Main Objective

To engineer a comprehensive unified assistive intelligence system that empowers caregivers by integrating causal stress modeling, privacy-safe monitoring, adaptive skill training, and hands-free documentation into a unified operational workflow.

1. **To predict and visualize workforce burnout dynamics using robust causal inference**
   - By applying co-teaching techniques to refine noisy stress data and detecting structural regime shifts in workload, allowing for personalized cluster analysis and clear visualization of potential intervention effects.

2. **To guarantee care recipients safety through privacy-preserved, continuous monitoring**
   - By processing real-time camera feeds to detect falls and anomalies without storing video, providing caregivers with timeline analysis of risk events and detailed feedback reports on care recipients activity patterns.

3. **To drive skill retention and communicative competence via adaptive technology**
   - By employing interactive video avatars within a personalized learning path that adapts to individual progress, ensuring high retention rates of sign language through real-time correction and gamified practice.

4. **To operationalize hands-free documentation and intelligent decision support**
   - By leveraging voice-activated logging to produce instant report summaries and utilizing domain-specific NLP to assist specific caregiver clusters when unsure, while ensuring robust security and personalization for senior staff records.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22178640	**AI Assisted Caregiver Workforce Deterioration Detection and Adaptive Workload Redistribution -** A core system component that tracks caregiver burnout over time, detects critical breakdown points where usual recovery fails, and explains the potential impact of alternative support actions, enabling earlier and more effective intervention in high-pressure care environments.	**Conceptualizing the Neuro-Symbolic Causal Architecture:** Defining the boundaries of the problem through the critical evaluation of existing literature regarding the dynamics of burnout in the caregiver workforce and the limitations of correlation-driven predictive models through Neuro-Symbolic approach and pinpoint the gaps in the existing frameworks of data science. **Engineering Robust Feature Pipelines for Multimodal Data**: Building data pre-processing and feature extraction pipelines for benchmark physiological and acoustic data. This involves applying signal processing algorithms for simulating sensor data streams and developing domain-specific features such as Heart Rate Variability (HRV) descriptors and spectral centroids that can adapt to poor-resource settings.	This component is novel because it fundamentally shifts stress monitoring from passive prediction to root-cause understanding. While existing AI algorithms only point out high levels of stress based on correlations, this model uses the concept of Temporal Causal Discovery to interpret the reasons for burnout symptoms and understands how factors like inadequate sleep in the past three days contribute to current feelings of fatigue. Most importantly, this solves the issue of using uncertain real-world data with its 'Co-teaching Deep Learning Framework.' Unlike most approaches, which require completely accurate data, this 'Co-teaching Deep Learning Framework' allows for two neural networks to learn together, helping them to mutually check and eliminate incorrect labels. The Neuro-Symbolic Methodology, in which rules defined

		Deriving Acoustic and Behavioral Biomarkers via Signal Processing: Applying advanced algorithms for audio signal processing to identify non-verbal vocal cues such as pitch, jitter, and shimmer from short audio logs recorded during high-demand care routines. This involves extracting structured behavioral biomarkers from unstructured caregiver voice data to serve as a non-invasive, secondary validation channel for detecting physiological stress and cognitive fatigue. **Architecting Noise-Resilient Deep Learning Models (Co-teaching) :** Developing a noise-tolerant deep learning framework that trains two deep learning models simultaneously to cross verify and remove noisy data samples. The framework utilizes the Co-teach training strategy to adaptively identify and learn from the highest-quality training data samples.	by experts are used to avoid illogic in predictions in data-driven models of AI. There is also prescriptive analysis capability by way of Counterfactual "What-If" Simulations, allowing managers to mathematically model whether a change in staff rostering can minimize stress levels before making the change. Ultimately, using Structural Regime Shift Detection, the system recognizes the point of failure in a caregiver's natural capacity for healing in a dynamic model of caregiver burnout.

		Modeling Personalized Causal Graphs and Structural Regime Shifts: Applying Causal Graph Discovery to optimize workforce allocation by mapping unique stress triggers for each caregiver. This involves building Personalized Causal Graphs to detect individual sensitivities and identifying Structural Regime Shifts the critical "breaking points" where a caregiver's ability to recover stops working (e.g., when sleep no longer reduces stress). This enables the system to recommend workload redistribution before permanent burnout occurs, moving from passive tracking to proactive prevention. **Implementing Neuro-Symbolic Counterfactual Inference Engines:** Building a logical reasoning engine that could support "What-If" simulations based on the learned structural causal models and focused on the mathematical analysis of the possible intervention effects.	

		Developing a Prescriptive 'Roster-Repair' Recommendation Engine: Moving on from the realm of simple monitoring, this involves the creation of an intelligent recommendation mechanism that takes the limited nature of a caregiver's resilience into consideration. Unlike the simple alert of "High Stress," the engine makes use of the principle of causal predictions to create optimal recommendations on schedule changes, such as a recommendation to switch a tired caregiver with a rested one.	
IT22057488	**Edge-Enabled Skeletal Movement Analysis for Predictive care recipients Safety Monitoring** - The component tracks care recipient's movement in real-time based on anonymized data of skeleton joints and removes every bit of video data recorded for anonymity and related privacy reasons. The	**Privacy-Preserving Movement Data Extraction :** The system will begin extracting only the skeletal positions of the joints from the live video feeds while discarding the video frames immediately. This means that the system will not be preserving any identifiable information from the video. The video information will be converted into numeric movement data that does not contain any identities. In terms of data science analysis, the analysis will be conducted on the non identifiable data.	The originality of the system consists in the fact that it is specifically designed to aid caregivers and not replace them or add to their workload. Unlike other systems that monitor care recipients through continuous video recording or mere alerts concerning the occurrence of a fall, the system can provide caregivers with early and informative warnings regarding the danger to the care recipients while at the same time safeguarding care recipients' privacy. Since the system is able to transform the images from the cameras into

	component alerts caregivers only when it detects movement instability in terms of minute changes in balance and ignores other movements that could pose no actual danger to the care recipient's stability.	**Modeling Care recipients Movement Over Time Using Graphs :** The movement of care recipients will be modeled using spatio-temporal graphs, where the human body's joints will be considered as a series of connected points, and their movement patterns will be analyzed. This will enable the system to recognize the way various parts of a human body move together. This is not possible through frame analysis. **Early Prediction of Falls and Risky Behaviors :** Rather than tracking only completed falls, machine learning algorithms will be able to recognize warning signs of dangerous events. From tracking past movement patterns, the system will be able to predict dangerous events before they even occur, allowing professionals to act accordingly. **Real-Time Monitoring Using Edge Devices:** For this purpose, the movement analysis algorithms will be executed on the local edge devices. With this approach, there will be instant	anonymous skeletal movements, caregivers can monitor the safety of the care recipients without necessarily watching or recording the actual images. Another innovative feature is the system's capability to identify early signs that might indicate instability and discomfort on the part of the care recipients. Caregivers get the capability to take preventive measures against falls and injuries. Moreover, the system can reduce unnecessary alerts that overburden the caregivers, allowing them to concentrate on actual situations that deserve attention. Through the use of skeleton replay animations, caregivers get the opportunity to take care of themselves.

		analysis of the movement data without any interruptions or internet connectivity issues. In addition, local data analysis ensures privacy is maintained in the care setting since the data is under local control. **Reducing False Alarms with Intelligent Filtering:** The system will also incorporate algorithms that recognize safe activity and actual threats to safety. The activities will also have to continue for a short duration before triggering an alarm to ensure that actual notice is not drawn to activities that may appear to pose threats to safety but that are harmless, like stretching or reaching for things. **Reliable Multi-Sensory Alert Notifications :** When a high-risk situation is identified, caregivers are notified through a variety of senses, including audio and continuous vibration. Not only are critical notifications acknowledged instantly, but the notification system is also designed so that caregivers are not	

		burdened with unnecessary alerts for urgent events. **Privacy-Safe Visual Confirmation Using Skeleton Replays :** In addition, every alert notification will include a skeleton animation replay of the care recipient's activity with a simple stick-figure representation. The purpose of this is to enable healthcare providers to quickly review what has occurred without requiring them to depend on video. Additionally, this data is based only on skeletal information to ensure care recipients anonymity while increasing the accuracy of alerts. **Long-Term Movement and Stress Pattern Analysis:** Over time, the system can use analysis of alert patterns to determine trends in increased activity at night, excessive instability, or other indicators of restlessness. All of these can aid in recognizing conditions. Not only can the system recognize when a fall has occurred but also aid in monitoring health conditions.	

IT22294098	**Context-Aware Intelligent System for Automated Extraction, Structuring, and Real-Time Documentation of Activities of Daily Living (ADL) from Multilingual Caregiver Voice Logs** –A context-aware system that transforms unstructured speech into precise, real-time recordings for better monitoring and care continuity by automatically extracting, organizing, and documenting Activities of Daily Living (ADL) from multilingual caregiver voice logs.	**Dialect and Multi-Language Speech Analysis** Model caregiver speech in Singlish, English, and regional accents. For accurate voice tracking, record colloquial language and culturally relevant idioms. Assure accuracy, accessibility, and inclusion in a variety of caregiving contexts. **Privacy-Preserving Speech-to-Text Optimization** To convert speech to text while protecting care recipients' privacy, implement and refine an offline ASR engine. Modify the system to include ADL phrases, medicine names, and words unique to caregiving. Assess performance in mobile, noisy, and accent-varying environments. **AI Auto-Editing, Context-Aware Speech Enhancement, and Real-Time Correction** Transcriptions can be automatically cleaned and improved by adding punctuation, capitalization, formatting, and eliminating fillers. Maintain documentation's domain-specific	This component is unique because it combines context-aware intelligence and sophisticated voice AI to create a smooth, automated platform that replaces traditional caring documents. It enables reliable transcription of caregiver instructions in real-world, noisy contexts by combining privacy-preserving offline processing with multilingual and accent-aware voice recognition. Real-time auto-editing, wake-word prompted ambient scribing, and domain-specific natural language processing that transforms unstructured speech into organized ADL records all improve usability .In order to provide proactive insights for care management, the system also integrates early risk detection, predictive analytics, and a care recipients digital twin for visual monitoring. Furthermore, it uses a 3D avatar to automatically generate structured daily reports and concise AI-spoken shift summaries. It incorporates context-aware guidance and historical log retrieval to minimize handover errors, guarantee

		meaning. Use real-time correction to ensure accurate, current ADL logs by allowing caregiver modifications to smoothly replace earlier entries. **Domain-Specific NLP and Recognition of Named Entities** Use a customized NER model to convert unstructured text into structured ADL records. Activities, symptoms, medication names, dosages, time references, and emotional cues should all be extracted. Verify outputs for dependability to facilitate reporting and analytics. **Wake-Word Based Ambient Scribing and Voice Shortcuts** Use wake-word detection to enable hands-free logging (e.g., "Hey CareLog"). To record caregiver voice without requiring device involvement, the recording will start and stop automatically. Boost adoption and usability while minimizing interference with caring duties.	continuity of care, and support data-driven clinical decision-making, resulting in an end-to-end intelligent caregiving solution that is both creative and useful.

		Predictive Care Insights & Early Risk Detection with Notifications Examine past ADL data to find trends that point to new dangers, such as poor sleep, decreased mobility, or skipped meals. Provide caregivers with real-time warnings and alerts so they can take preventative action. Improve care recipients safety and make data-driven decisions possible. **Care recipients Digital Twin State Modeling and Visual UI** Using ADL data, create a 3D digital twin to show the functional and health state of the care recipients. For intuitive comprehension, graphically depict weariness, nutrition, movement, and skin risk. Promote quicker clinical decision-making and improve situational awareness. **Automated Shift Handover and Intelligent Care Reporting** Using a 3D avatar, automatically produce structured daily reports and 30-second AI-spoken shift summaries. Incorporate context-aware instructions and historical log retrieval. Improve	

		continuity of care, lower handover errors, and encourage well-informed decision-making.	
IT22289384	**Personalized Sign Language Tutoring System for Caregivers-** Caregiver-oriented learning module that combines computer vision, spatiotemporal gesture analysis, and real-time 3D avatar feedback to teach sign language effectively.	**Creating a Computer Vision pipeline for Kinematic Attribute Extraction** Developing a strong computer vision system that monitors hand joints to establish a digital reference of the caregiver's actions. This guarantees that the system can effectively interpret hand gestures irrespective of the caregiver's hand size or their natural movement speed, establishing a basis for a genuinely tailored evaluation. **Formulating Temporal and spatial variations Analysis** Developing a comparison algorithm based on mathematical evaluation, using time alignment, the system is designed to adjust to the user's speed and not hinder the user if the user signs slowly during the early stages of learning the basics of sign language.	This component's novelty is its comprehensive, caregiver-oriented learning architecture, which combines gesture capture, spatiotemporal evaluation, interpretable feedback, and retention analysis into an integrated personalized system. Instead of using fixed-threshold scoring or static demonstrations like videos and images, the system uses precise mathematical modelling to create a caregiver-specific kinematic representation from hand-joint motion data and compare it to expert benchmarks. This design specifically separates movement precision from execution speed, allowing for fair evaluation for inexperienced students and considering the irregular practice patterns common in caregiving environments. By offering direct visual connection between the caregiver's hand movements and the reference

		Generating Personalized XAI Feedback via Rule-Based Logic Using Rule-Based Logic to Provide Individualized XAI (explainable) Feedback. Implementing a Virtual Coach layer where complex error messages will be converted into simple instructions. The system mimics a one-on-one tutoring experience by using rule-based logic to offer customized guidance (e.g.: Relax your thumb slightly) that addresses the precise error the caretaker makes. **Personalized Curriculum Adaptation via Probabilistic Knowledge Modeling** It creates a sequencing platform that is intelligent to adapt to the learning speed of the caregiver. Based on past mistakes made in signing, the program is able to determine which signs a user has mastered, and which signs need more attention and adjusts the learning accordingly to devote more time to learning difficult concepts.	gesture, a real-time 3D avatar interface enhances learning even further and encourages effective muscle memory building and effortless self-correction.

		Real-Time 3D Avatar-Based Visual Feedback System for Gesture Learning Creation of an immersive user interface that projects the movements of the caregiver's hands onto a 3D avatar in real time. This is ideal for visual learners in that it allows the caregiver to view themselves side by side with the tutor. This feedback channel is imperative in creating muscle memory for retention. **Designing Data-Driven Gamification for Skill Retention.** Developing a behavioral reward system that transforms tedious practice into a fun game. The system uses visual rewards and progress tracking to encourage daily practice through a scoring mechanism based on improvement levels. This gamified strategy is intended to increase motivation and guarantee that hectic caregivers remember the hand gestures over time. **Statistical Analysis of Learning Retention and System Adaptivity** Creating a formal analysis to demonstrate that this personalized,	

		gamified approach works better than traditional video learning. This involves evaluating user data to determine long-term retention rates (signs remembered after two weeks) and confirming that adaptive pacing increases caretakers' involvement and reduces irritation.	

AI USAGE DISCLOSURE

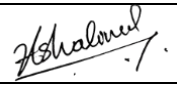
AI tools were used in a limited and supportive capacity during the preparation of this topic assessment report. ChatGPT was utilized primarily to enhance language clarity, refine sentence structure, and assist in the organization of ideas. The purpose of using these tools was to improve the readability and presentation of the document, while ensuring that all core research concepts, problem formulation, and methodological decisions were developed independently by individual members.

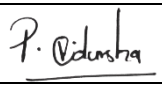
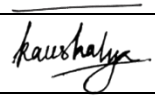
The extent of AI usage was strictly limited to minor writing assistance and idea refinement and did not involve generating full sections of the topic assessment report or replacing original intellectual contribution. All AI-assisted outputs were critically reviewed, modified, and validated by the authors to ensure accuracy, relevance, and alignment with academic standards. Validation methods included cross-checking information with credible academic sources, verifying technical correctness, and ensuring consistency with the research objectives. Full responsibility for the content presented in this proposal is retained by the authors.

We hereby declare that:

1. We have read and understood the AI Usage Policy for the Research Project (IT4010).
2. We acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. We agree to use AI tools responsibly, ethically, and transparently.
4. We will clearly disclose any use of AI tools in my submissions.
5. We understand that I bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. We confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
7. We understand that failure to comply with this policy may result in academic penalties.

Student Details :

Full Name	Registration Number	Signature
Dayana Priyadharshani Kumar	IT22178640	
Harishalinee Elangovan	IT22057488	

Vidursha Prabagaran	IT22294098	
Kaushalya Nagenthraraja	IT22289384	

1. **This part is to be filled by the Supervisor and the Co-supervisor of the Project.**

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Rercommended.

	Title	First Name	Last Name	Signature
Supervisor	Prof.	Samantha	Thelijjagoda	
Co-Supervisor	Mr.	Samadhi	Rathnayake	
External Supervisor	Mr.	Velu	Sivakumar	

Summary of external supervisor's (if any) experience and expertise:

Mr. Velu Sivakumar is a certified caregiver with over 2.5 years of professional experience, providing dedicated care in the UK. He holds a caregiver certification from IIHS Sri Lanka and, as a senior caregiver, oversees care operations while training and mentoring new caregivers, demonstrating his expertise and commitment to high standards of care.